Complete AI Engineer Training: Python, NLP, Transformers, LLMs, LangChain, Hugging Face, APIs

Section 1: Intro to AI Module: Getting Started

What does the course cover

AI Engineer Bootcamp designed to take learners from basic concepts to advanced, real-world AI applications through a structured, step-by-step approach. The program starts with foundational AI concepts and Python programming, then moves into natural language processing (NLP), large language models (LLMs), transformer architectures, and practical tools such as LangChain and vector databases like Pinecone.

The instructor, Ned, founder of 365 and an experienced Udemy educator, introduces the course team—data science and computer science experts from leading European universities. Together, they have created a comprehensive and practical curriculum aimed at professional growth in the AI field.

The course covers:

- Fundamentals of artificial intelligence, its branches, tools, and career paths
- Python programming from basics to practical use cases like APIs and web scraping
- NLP as the foundation of generative AI
- Large language models and transformers
- Integrating LLMs into web applications using LangChain
- A final hands-on case study that consolidates all learned concepts

Learners are advised to follow the course sequentially, as each lecture builds on previous material. The program includes downloadable resources such as notes, PDFs, exercises, and notebooks to reinforce learning. Completing exercises is strongly encouraged to strengthen practical skills.

Students who complete 50% of the course and announce it in the Q&A section receive a bonus: exclusive access to a ChatGPT for Data Science course. The instructors actively encourage participation through questions and discussions in the Q&A section and express enthusiasm for learners to begin the journey.

Natural vs Artificial Intelligence

Natural intelligence is demonstrated through activities such as driving, solving complex mathematical problems, writing poetry, and mastering music. These abilities are not present at birth and are developed over time through learning. Learning is possible because the human brain can observe, process, and apply large amounts of information.

Intelligence is defined as the ability to acquire and apply knowledge and skills. Human intelligence has led to technological innovations that far surpass the tools available in earlier centuries.

Humans are effective tool builders and have continuously created tools to increase productivity. One of the most significant inventions was the Gutenberg printing press, developed in 1440, which transformed the way knowledge is shared. Despite its importance, the printing press cannot be considered an intelligent machine because it operates with fixed rules and cannot learn or adapt.

The concept of intelligent machines emerged much later. Inspired by human intelligence and brain processes, computer scientists sought to enable machines to acquire and apply knowledge. This effort led to the development of artificial intelligence.

Brief history of AI

Artificial Intelligence (AI) has evolved through several important scientific and technological breakthroughs over the past decades.

In **1950**, British mathematician and computer scientist **Alan Turing** posed the question *"Can machines think?"* in a scientific paper. His major contribution was shifting the discussion from philosophy to experimentation. He proposed the **Turing Test**, a practical method to evaluate machine intelligence. In this test, if a human interrogator cannot distinguish between a machine and a human through conversation, the machine is considered intelligent. This marked a foundational moment in AI research.

In **1956**, the term **Artificial Intelligence** was officially coined at the **Dartmouth Conference**, where AI became a formal academic field. Researchers explored whether machines could simulate aspects of human intelligence using concepts from neural networks, computation theory, and automata theory. From the beginning, AI research was inspired by how human intelligence works.

However, progress was slow due to limited computing power and lack of data. The **1960s and 1970s**, known as the **AI Winter**, saw reduced funding and interest in AI because expectations were not met.

A major breakthrough occurred in **1997** when **IBM's Deep Blue** defeated world chess champion **Garry Kasparov**, demonstrating the potential of machine intelligence. In the late 1990s and early 2000s, rapid growth in internet usage and computing power created the necessary conditions for AI advancement. Massive amounts of digital data became available, and computational capabilities increased dramatically.

In **2006**, **Geoffrey Hinton** revived interest in neural networks through **deep learning**. Deep learning models, inspired by the human brain, require large datasets and high computational power—resources that were previously unavailable.

In **2011**, **IBM Watson** won the quiz show *Jeopardy!*, showcasing significant progress in **natural language processing**, enabling AI systems to understand and process human language.

In **2012**, researchers from **Stanford and Google**, including **Jeff Dean and Andrew Ng**, published influential work on large-scale unsupervised learning. Their deep neural network system successfully recognized complex visual patterns, such as identifying cats, marking progress in multilayer neural networks.

In **2017**, the **Google Brain** team introduced **Transformers**, a breakthrough model using self-attention mechanisms that greatly improved how machines process sequences like text.

In **2018**, **OpenAI** released the first version of **GPT (Generative Pre-trained Transformer)**, using Transformers to create large language models (LLMs). This innovation led to rapid improvements in generative AI and eventually to user-friendly systems like **ChatGPT**.

By **late 2022**, generative AI became widely accessible, marking a new era in artificial intelligence. Although the concepts may initially seem complex, they become clearer as learning progresses.

Demystifying AI, Data science, Machine Learning, and Deep Learning

The terms **Artificial Intelligence (AI)**, **Machine Learning (ML)**, and **Data Science** are closely related but have distinct meanings, and they should not be used interchangeably.

Artificial Intelligence is a broad discipline focused on making machines intelligent. Its goal is to enable systems to learn, adapt, and acquire new skills in a manner similar to human intelligence. AI encompasses several subfields, one of the most important being machine learning.

Machine Learning is a major subfield of AI that uses data to predict outcomes or make decisions. ML models learn complex patterns and dependencies in data that go beyond simple correlations. Similar to a mathematical function, input data is processed by a model to produce an output.
For example, machine learning algorithms can analyze a user's movie-watching history to recommend new movies or evaluate financial transaction data to generate a credit score that predicts loan repayment behavior.

Data Science has a strong intersection with AI and machine learning but extends beyond them. Data scientists commonly work with machine learning algorithms, but they also use traditional statistical techniques such as data visualization, statistical inference, and exploratory data analysis. Valuable insights can often be extracted from data without relying on machine learning.

Data science integrates AI, machine learning, mathematics, statistics, and visualization techniques to derive business value from data. For instance, a data scientist may build a predictive model to forecast future customer orders or analyze and visualize customer behavior to uncover meaningful trends and insights.

Weak vs Strong AI

Artificial intelligence systems vary in capability and scope. Some AI systems are designed to perform a single, well-defined task. This form of intelligence is known as **Narrow AI**. Such systems can only do what they are specifically trained for and do not possess general reasoning abilities. Narrow AI is already widely used in daily life and business applications, such as recommendation systems that suggest movies based on viewing history.

More advanced systems fall under **Semi-Strong AI**. A notable example emerged in **2022** with the release of **ChatGPT and ChatGPT-3.5** by OpenAI. These systems demonstrated the ability to perform a wide range of tasks rather than a single specialized function. They can generate human-like responses, write jokes, proofread text, recommend actions, analyze and create images, generate programming formulas, and solve mathematical problems. Their conversational ability often makes them indistinguishable from humans, aligning closely with **Alan Turing's imitation game**, also known as the **Turing Test**. At this level, AI systems can be considered semi-strong due to their versatility and breadth of skills.

The next and most advanced stage is **Artificial General Intelligence (AGI)**, also referred to as **Strong AI**. AGI has not yet been achieved. According to **Sam Altman**, CEO of OpenAI, AGI would be the most powerful technology humanity has ever created. Such an AI would surpass human capabilities across many tasks and possess true general intelligence. Many researchers believe that a key requirement for AGI is the ability to create new scientific knowledge independently by synthesizing existing information to generate original discoveries and ideas.

Opinions differ on how close humanity is to achieving AGI, but some experts believe it could emerge in the near future. As progress continues toward this goal, it raises profound questions about the potential impact and ethical responsibilities associated with creating machines that can think, learn, and innovate more effectively than humans.

Section 2: Intro to AI Module: Data is essential for building AI

Structured vs unstructured data

Data is the core requirement for building any AI model.

There are two main categories of data: **structured** and **unstructured**.

Structured data is neatly organized into rows and columns (e.g., Excel sheets, databases), making it easy to store and analyze.

Unstructured data includes text, images, videos, and audio, which do not follow a fixed tabular format.

Approximately **80–90% of all global data is unstructured**.

In the past, structured data was seen as more valuable because it was simpler to process.

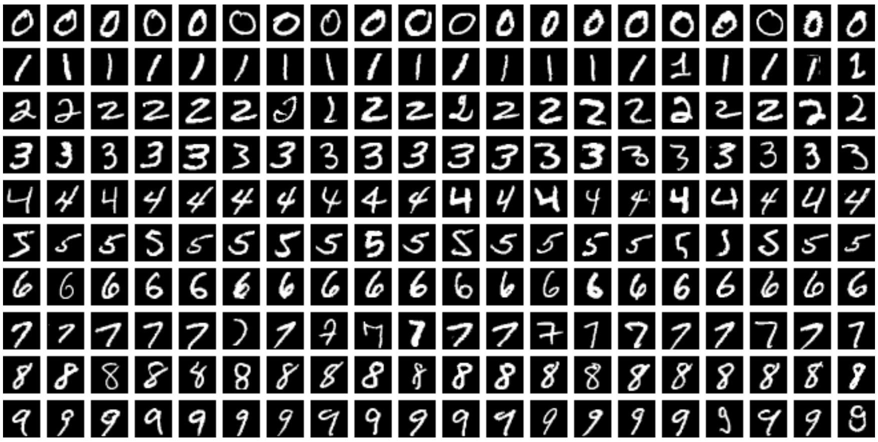
Modern AI techniques now enable valuable insights to be extracted from unstructured data.

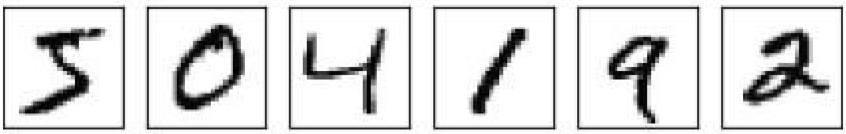
Companies such as **Google and Meta** are leaders in leveraging unstructured data at scale.

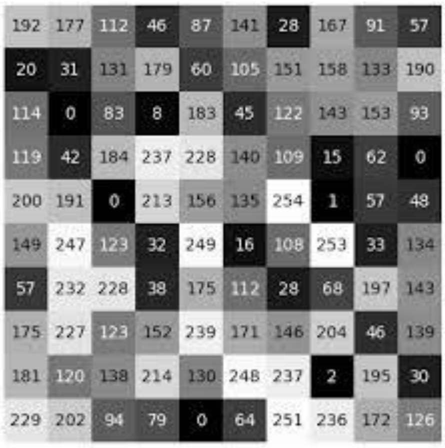
Analyzing unstructured data allows businesses to gain deep insights from photos, videos, emails, and messages.

The next lesson focuses on **how AI systems interpret and learn from unstructured data**.

How we collect data







Summary: MNIST and How Computers Learn from Data

- The **MNIST dataset** is considered the *“hello world”* of machine learning and is widely used by beginners.
- It contains about **70,000 images** of handwritten digits (0–9), each sized **28×28 pixels** in grayscale.
- The goal is to train a machine learning model to **recognize digits**, even though each digit looks different due to variations in human handwriting.

How computers understand images

- Every image is made up of **pixels**, and each pixel has a grayscale value from **0 to 255**
    - 0 → white
    - 255 → black
- Computers store and process these pixel values in **binary form (0s and 1s)**.
- As a result, an image becomes a **numerical sequence**, not a visual object, from the computer’s perspective.
- Different handwritten versions of the same digit (like “3”) look different to humans but produce **similar numerical patterns** for the computer.

Learning with MNIST

- Machine learning algorithms learn by identifying **patterns and similarities** in these numerical representations.
- Through training, the model learns to distinguish between digits such as **3, 6, and 9**, based on their pixel-value patterns.
- MNIST demonstrates how **real-world information can be converted into data** that machines can process.

Beyond images

- Videos are sequences of images (frames), each made of pixels.
- Audio and speech can also be represented as numerical data.
- Ultimately, **all digital information is reduced to zeros and ones**, giving computers a structured way to learn.

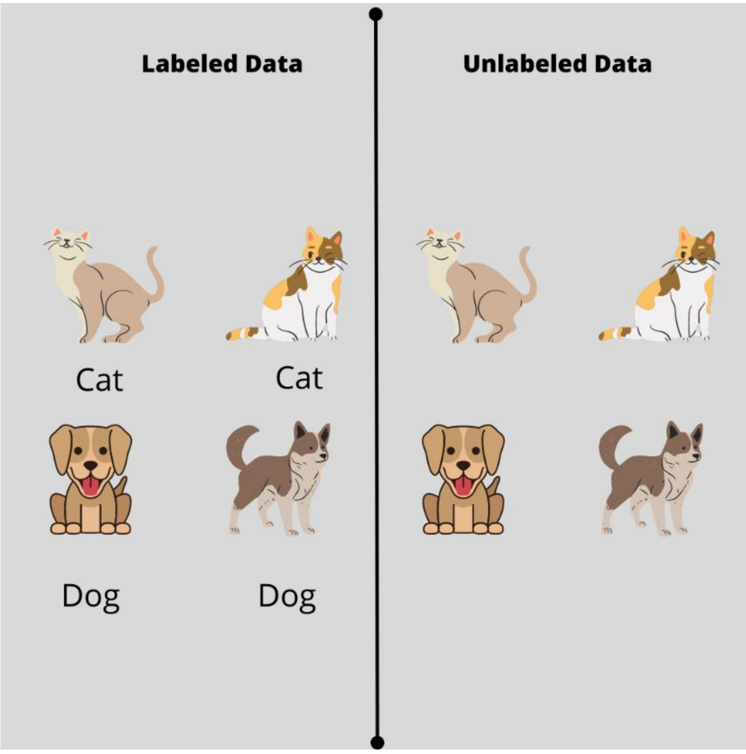
Human intelligence vs AI

- Humans naturally process massive amounts of sensory data simultaneously (sight, sound, touch, taste).
- AI researchers aim to replicate this capability using data from **sensors, images, videos, audio, text, social media, satellites, and web activity**.
- Data is collected through **APIs, web scraping, and big data analytics** to train and improve AI models.

Key principle

- *"Garbage in, garbage out"* — the **quality of data** directly determines the **quality of AI model outputs**.
- High-quality data is essential for building effective and reliable artificial intelligence systems.

Labelled and unlabelled data



Labeled vs Unlabeled Data in AI

- There are **two primary approaches to preparing data** for AI models: **labeled data** and **unlabeled data**.

Labeled data

- In labeled datasets, each data item is **manually tagged** with the correct category or outcome.
- Example: Reviewing **10,000 animal photos** and labeling each as *dog* or *not a dog*.
- The same approach applies to **text, audio, video**, and other data types (e.g., labeling YouTube comments as positive, negative, or neutral).
- Although **time-consuming and expensive**, labeled data produces **high-quality training data**.
- Models trained on labeled data are generally **more accurate, reliable, and effective in real-world applications**.

- The key trade-off is **higher cost and effort versus better performance**.

Unlabeled data

- Unlabeled datasets contain data **without predefined categories**.
- Modern machine learning techniques can learn patterns directly from **unstructured data** such as images, text, video, and audio.
- Example: Feeding the same 10,000 photos into a model **without labeling them beforehand**.
- The model is left to **discover patterns on its own**, reducing manual effort and cost.

Key takeaway

- You should clearly understand the **difference between labeled and unlabeled datasets**.
- Choosing between them involves balancing **data preparation effort, cost, and model performance**.

Metadata: Data that describes data

Data Growth and the Role of Metadata in AI

- The rapid **advancement of AI** is largely driven by the **digitalization of everyday life**.
- Growth in **online shopping, smartphones, cameras, social media, sensors, and IoT devices** has caused an **exponential increase in data generation**.
- Today's data is not only more abundant but also **higher in quality**, as seen in the improvement of modern smartphone photography compared to older devices.
- This explosion of **high-quality data** has strongly accelerated AI development.

Challenge of modern data

- Much of the data generated today is **unstructured** and **too large to be manually labeled**.
- Making sense of this massive volume of data requires additional context.

Metadata

- **Metadata is data about data**.
- It provides descriptive information such as **file type, author, creation date, usage details, and file size**.
- Example: In an image dataset (e.g., dog photos), metadata may include the photo name, capture date, photographer, and file size.

Importance of metadata

- Metadata is essential for **organizing, managing, and understanding data**.
- It is valuable across **structured and unstructured**, **labeled and unlabeled** datasets.
- In **large-scale datasets and complex AI systems**, metadata plays a critical role in enabling efficient data processing and analysis.